

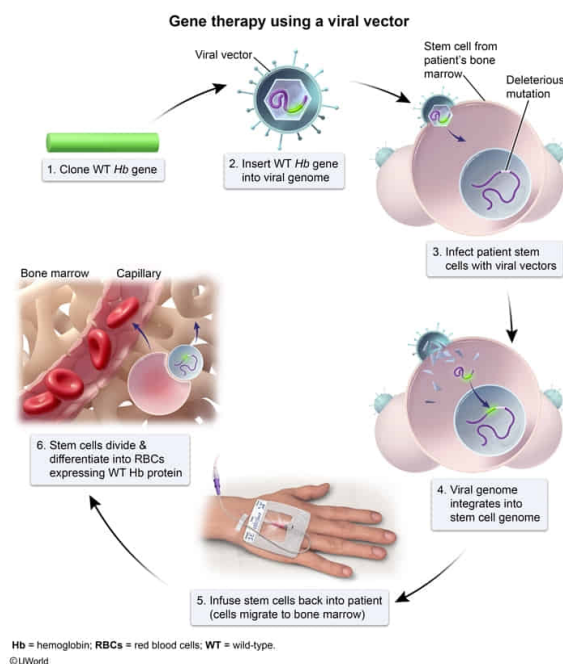
Several techniques have been developed to treat blood disorders caused by genetic mutations. Which of the following methods would be most effective in the treatment of a patient homozygous for a deleterious mutation in a gene coding for a subunit of hemoglobin?

- ☐ A. Replacement of the mutant gene with a functional copy in red blood cells [15%]
- ☐ B. Treatment with a drug that increases red blood cell production [2%]
- ✓ ☒ C. Introduction and activation of a functional hemoglobin gene in bone marrow stem cells [76%]
- ☐ D. Single blood transfusion from a patient homozygous for the wild-type hemoglobin gene [5%]

Correct

74% answered correctly

Explanation:



Hemoglobin is an oxygen-binding protein found within **red blood cells** (RBCs). Oxygen diffuses into the bloodstream via **gas exchange** in the lungs, binds hemoglobin, and is transported through the **circulatory system** to tissues throughout the body. Deleterious mutations affecting hemoglobin genes may negatively impact hemoglobin function, impairing oxygen transport.

One method of treating genetic mutations is **gene therapy**, a technique in which a functional gene is introduced into a patient's cells to replace a mutant gene. The new gene can then be **transcribed and translated** into functional proteins. A common gene therapy method uses **retroviral** vectors to insert functional gene copies into the stem cells of patients exhibiting specific genetic mutations.

In this scenario, a patient is homozygous for a deleterious mutation in a gene coding for a hemoglobin subunit. Expression of this mutant gene would likely produce a portion of the

hemoglobin protein with impaired function. Because RBCs are *produced* from stem cells in the **bone marrow**, the most effective treatment for this patient would be gene therapy to **introduce** and **activate** a **functional hemoglobin gene** in **bone marrow stem cells**. This would likely lead to the production of RBCs that contain hemoglobin with *normal* oxygen-binding abilities.

(Choice A) The mutant gene could *not* be replaced by a functional copy in mature RBCs because mature RBCs *lack* nuclei and do *not* contain DNA.

(Choice B) Treatment with a drug that increases RBC production would *increase* the differentiation of bone marrow stem cells into mature RBCs. However, because this treatment does *not* introduce a functional copy of the hemoglobin gene, these cells would still express the *mutation* as well as functionally impaired hemoglobin.

(Choice D) A blood transfusion from a patient homozygous for the wild-type hemoglobin gene may introduce RBCs expressing functional hemoglobin into the patient's bloodstream. However, all aging RBCs are eventually destroyed by **phagocytic** cells (ie, macrophages) in the spleen. Because RBCs with wild-type hemoglobin would eventually be destroyed, the patient would require *multiple* blood transfusions to maintain a population of healthy RBCs.

Educational objective:

Gene therapy is a technique in which a functional copy of a gene is introduced (eg, via retroviral vectors) into the cells of a patient with a mutation in that gene.

Time spent:0 sec

Subject:
Biology

QID:402601

System:
Circulation and Respiration